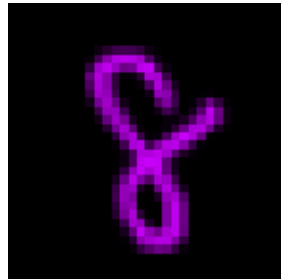
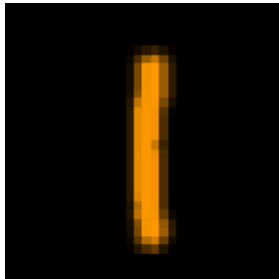
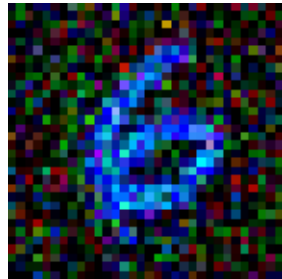
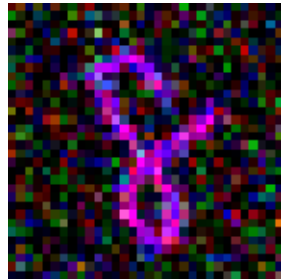
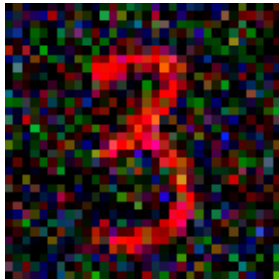
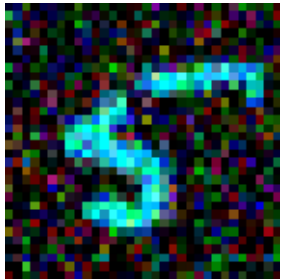
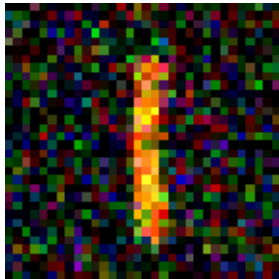


Huber Noise (alpha=1.0, scale=0.5)

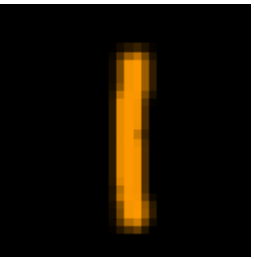
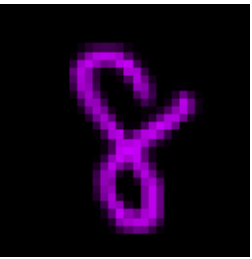
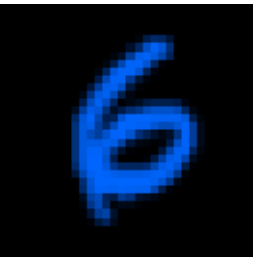
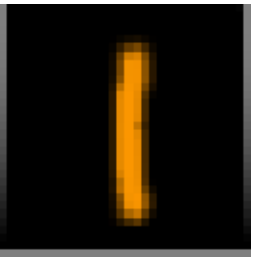
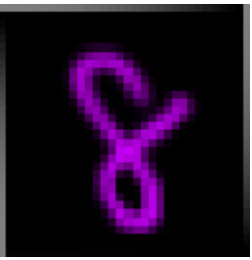
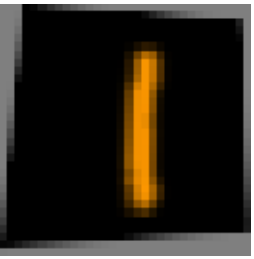
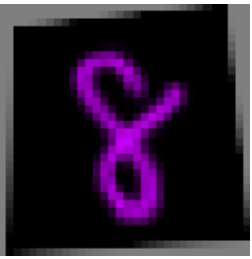
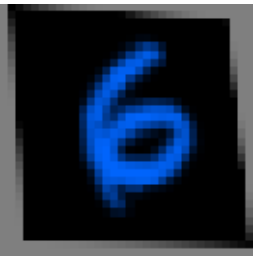
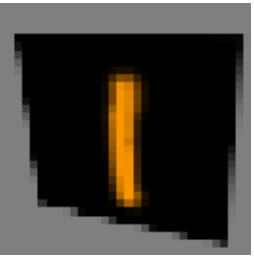
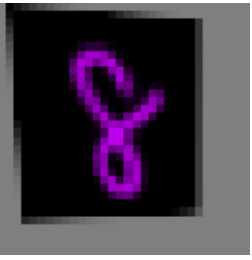
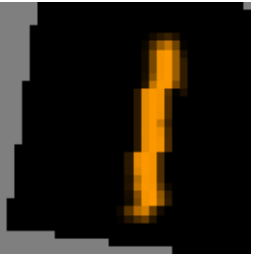
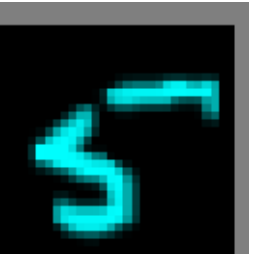
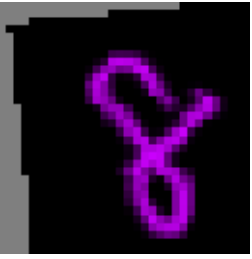
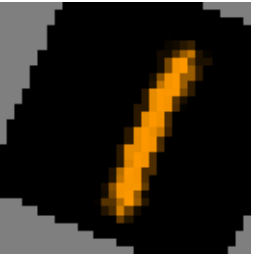
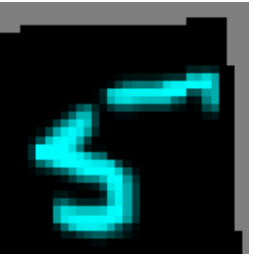
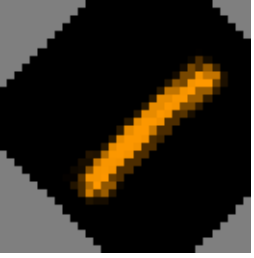
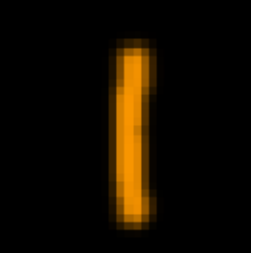
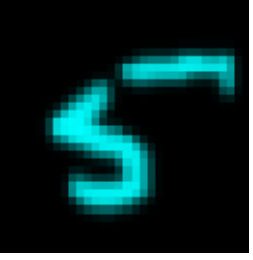
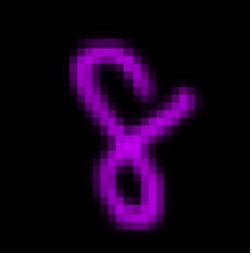
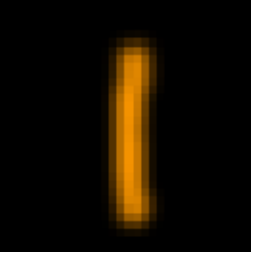
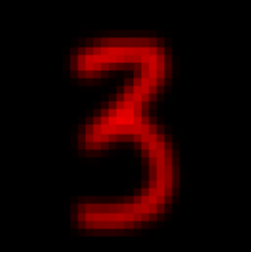
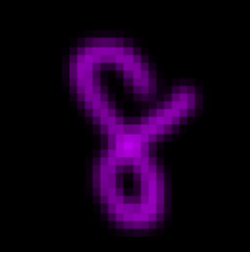
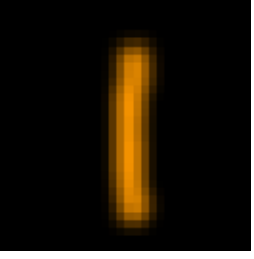
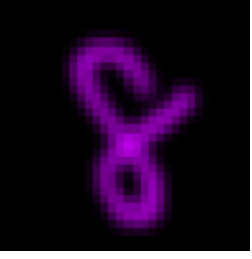
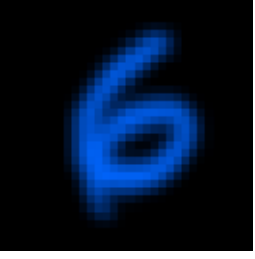
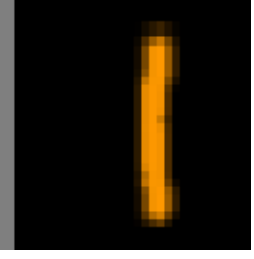
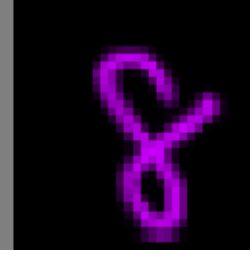
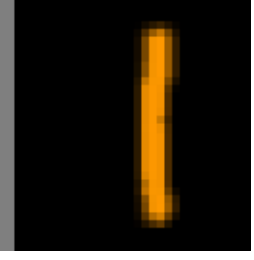
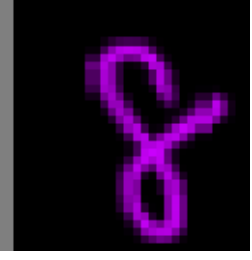
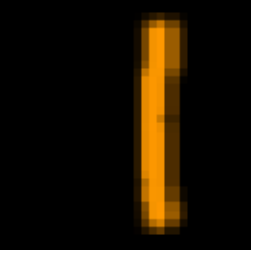
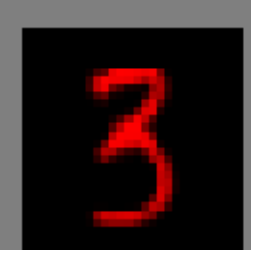
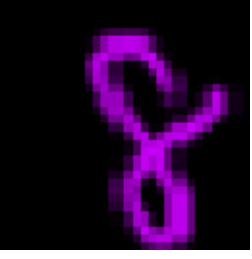
Original



Huber



Camera Shifts Visualization

Original					
Angle (low)					
Angle (med)					
Angle (high)					
Rotation (low)					
Rotation (med)					
Rotation (high)					
Blur (low)					
Blur (med)					
Blur (high)					
Zoom (low)					
Zoom (med)					
Zoom (high)					

Huber Noise - Relative L2 (%)

Method	Alpha=0.0	Alpha=1.0
Abs-LiNGAM (Noisy)	6637.66 ± 945.80 *	7470.00 ± 947.73 *
Abs-LiNGAM (Perfect)	107569.20 ± 13304.38 *	115113.65 ± 13200.41 *
BaryCA	3724666.76 ± 425740.41 *	3767850.68 ± 426066.52 *
DiRoCA (eps_20_delta_0.0)	** 427.53 ± 224.89 **	** 506.04 ± 225.94 **
DiRoCA (eps_40_delta_0.0)	1330.93 ± 391.43 *	1605.06 ± 388.32 *
DiRoCA (eps_61.686502153383906_delta_0.0)	3672.23 ± 1004.72 *	4111.68 ± 988.48 *
DiRoCA (eps_8.0_delta_0.0)	799.27 ± 449.95	896.83 ± 450.13
GradCA	5001759.36 ± 314549.77 *	5066962.76 ± 319233.05 *

** Best Method | * Significantly worse (p<0.05) than best

Camera Shift: Angle - Relative L2 (%)

DiRoCA	Method	Low	Med	High
	Abs-LiNGAM (Noisy)	4823.68 ± 540.03 *	3008.25 ± 423.39 *	2083.08 ± 372.63 *
	Abs-LiNGAM (Perfect)	65925.76 ± 4553.79 *	26989.24 ± 1755.84 *	9258.19 ± 935.12 *
	BaryCA	3072307.15 ± 279563.77 *	1112570.43 ± 106294.09 *	277881.10 ± 21188.63 *
	DiRoCA (eps_20_delta_0.0)	** 419.95 ± 236.29 **	** 390.36 ± 212.15 **	** 276.53 ± 112.12 **
	DiRoCA (eps_40_delta_0.0)	1192.21 ± 327.25 *	908.12 ± 187.95 *	522.50 ± 83.10 *
	DiRoCA (eps_61.686502153383906_delta_0.0)	3011.35 ± 772.41 *	1971.20 ± 429.24 *	1045.77 ± 195.94 *
	DiRoCA (eps_8.0_delta_0.0)	720.10 ± 349.36	794.88 ± 346.21	803.14 ± 293.62 *
	GradCA	2622784.18 ± 152353.79 *	888148.82 ± 72272.75 *	234263.81 ± 22296.70 *

Camera Shift: Rotation - Relative L2 (%)

Method		Low	Med	High
Abs-LiNGAM (Noisy)		4572.44 ± 583.12 *	4060.26 ± 547.59 *	3697.27 ± 536.54 *
Abs-LiNGAM (Perfect)		63600.29 ± 5360.62 *	50026.47 ± 4748.59 *	41327.62 ± 4159.11 *
BaryCA		2135130.10 ± 204889.69 *	1507062.73 ± 142498.19 *	1086491.09 ± 100974.65 *
DiRoCA	(eps_20_delta_0.0)	** 405.67 ± 219.79 **	** 394.24 ± 227.35 **	** 382.14 ± 231.57 **
	(eps_40_delta_0.0)	1089.11 ± 274.11 *	1009.22 ± 270.56 *	938.29 ± 263.20 *
	(eps_61.686502153383906_delta_0.0)	2670.54 ± 627.02 *	2371.50 ± 564.66 *	2124.07 ± 492.47 *
	(eps_8.0_delta_0.0)	787.86 ± 380.81	774.65 ± 361.33	771.81 ± 339.90
	GradCA	2444221.96 ± 169432.67 *	1826286.89 ± 131890.72 *	1371763.77 ± 105841.04 *

Camera Shift: Blur - Relative L2 (%)

Method		Low	Med	High
Abs-LiNGAM (Noisy)		6610.48 ± 945.59 *	6577.70 ± 945.16 *	6570.57 ± 945.03 *
Abs-LiNGAM (Perfect)		107492.64 ± 13292.10 *	107413.25 ± 13273.12 *	107397.76 ± 13268.12 *
BaryCA		3723919.71 ± 425516.59 *	3722771.51 ± 425179.67 *	3722465.66 ± 425092.15 *
DiRoCA	(eps_20_delta_0.0)	** 426.01 ± 224.29 **	** 424.31 ± 223.37 **	** 423.96 ± 223.12 **
	(eps_40_delta_0.0)	1324.21 ± 390.93 *	1316.42 ± 390.16 *	1314.78 ± 389.96 *
	(eps_61.686502153383906_delta_0.0)	3657.86 ± 1004.39 *	3639.65 ± 1003.76 *	3635.48 ± 1003.56 *
	(eps_8.0_delta_0.0)	796.99 ± 450.11	794.39 ± 450.36	793.87 ± 450.43
	GradCA	5001145.87 ± 314546.35 *	5000186.73 ± 314545.28 *	4999929.10 ± 314545.81 *

Camera Shift: Zoom - Relative L2 (%)

Method		Low	Med	High
DiRoCA	Abs-LiNGAM (Noisy)	4884.84 ± 565.08 *	4830.61 ± 570.40 *	4749.12 ± 552.30 *
	Abs-LiNGAM (Perfect)	70408.59 ± 5828.87 *	69351.71 ± 5864.97 *	66482.56 ± 5867.03 *
	BaryCA	2326371.58 ± 220738.17 *	2309686.87 ± 217797.25 *	2209107.49 ± 210481.51 *
	DiRoCA (eps_20_delta_0.0)	** 407.19 ± 218.08 **	** 402.59 ± 213.27 **	** 384.43 ± 193.76 **
	DiRoCA (eps_40_delta_0.0)	1101.15 ± 271.17 *	1087.83 ± 267.07 *	1027.51 ± 249.47 *
	DiRoCA (eps_61.686502153383906_delta_0.0)	2744.94 ± 642.12 *	2705.15 ± 619.24 *	2573.44 ± 595.18 *
	DiRoCA (eps_8.0_delta_0.0)	793.40 ± 383.25	793.18 ± 382.19	791.99 ± 375.67
	GradCA	2650430.28 ± 186616.93 *	2651583.62 ± 180042.69 *	2667782.49 ± 169276.88 *